802.15.4 Dataset

Authors

Adnan Iqbal and Khurram Shahzad

Release date

Feb 21, 2008

Measurement start/end date

Start: April 2007

End: June 2007

Measurement environment: some background info

The dataset has been collect at NUST School of Electrical Engineering and Computer Science, Rawalpindi. The dataset is collected to investigate bit-error process of 802.15.4 low data rate networks.

Measured network (technical description)

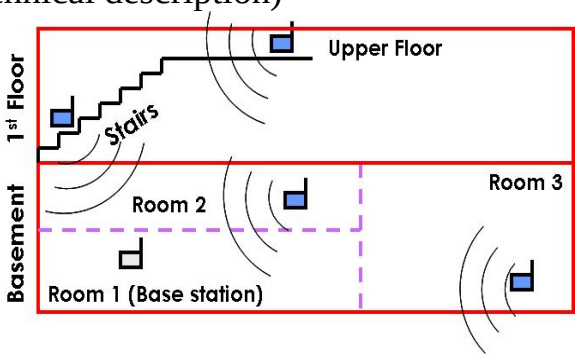


Figure 1: 802.15.4 Data collection setup

These Traces were collected in four different setups. Each setup is characterized by the distance from the base station and obstructions between sender and the base station. In each experiment, one sender transmitted data to the base station and the other senders were inactive. Distance between a sender and the base station varied from 5 to 12 meters. The senders transmitted fixed-sized 20-byte frames at a rate of 10 frames per second. In each setup, six or more traces were collected. The average number of frames per trace was approximately 31,000.

Setups used to collect traces are shown in Figure 1. These setups are named based on their geographical location. We observed maximum bit-error rate for setup named Room 3. The reason of high bit-error rate is the presence of concrete wall and longer distance.

Collection methodology

Hardware:

Crossbow MicaZ motes data rate 250kbps

Crossbow MIB510/MIB600 Programming Board

Software:

TinyOS 1.x in Cygwin

We used Crossbow's MicaZ motes and TinyOS (operating system) to collect residual bit-error traces. To program MicaZ motes the Crossbow MIB510 Serial Programming board and Crossbow MIB600 Ethernet Programming board were used. Same programming boards were later used as base-station to capture the traffic generated by programmed MicaZ motes.

TinyOS was modified to forward all received packets at base-station to upper layers (regardless of being received in error at MAC layer). To forward packets from base-station board to attached computer, we used Listen utility provided by TinyOS. However, this utility was also modified to retain erroneous packets.

Hole/error/limitations

The data set does not have any information about the packets which have been totally lost. Only those packets are captured which are received completely (although may be in error).

Anonymization technique

Not Applicable

Data format

The dataset consists of many files. Each file represents one complete trace. Each file has one frame per line. Frame contents are hexadecimal values. First eleven bytes are header bytes whereas remaining 20 bytes are data bytes. The data byte "AA" represents no error whereas any other value represents error in the received byte. Actual bit in error can be computed by XORing received byte to "AA".

802.11 Dataset

Authors

Syed Ali Khayam and Yong ju Cho

Release date

July 08, 2008

Measurement start/end date

Start: June 2006

End: July 2006

Measurement environment

These traces are collected at Wireless and Video (WAVES) Lab at Michigan State University (MSU). The basic purpose of collecting these traces was to investigate bit-error process at MAC layer of 802.11 traces and to study the relationship between Signal to Noise Ratio (SNR) at physical layer and bit-errors at MAC layer.

Measured network

For this study, we consider three different setups to encompass home, office and university settings. Two of these setups are shown in Figure 2. In Setup A, five wireless receivers were used to simultaneously collect error traces on an 802.11b WLAN. One receiver was placed within clear line-of-sight (LoS) of the access point (AP), while the remaining four receivers were placed at different locations in a room across the hallway. In setup B six receivers were used to simultaneously collect error traces. Three receivers were placed in a room across the hallway, while three receivers were placed (at an extreme edge of the network) in a room 100 feet down the hallway.

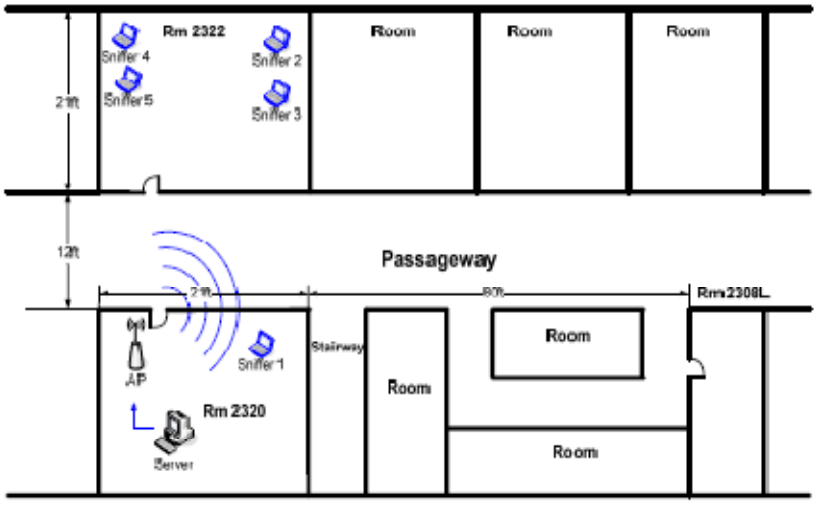
A wired sender was used to send multicast packets with a predetermined payload on the wireless LAN; multicasting disabled MAC layer retransmissions. Each experiment comprised of one million packets with a payload of 1, 000 bytes each. At the physical layer, the auto rate selection feature of the AP was disabled and for each experiment the AP was forced to transmit at a fixed data rate. Each trace collection experiment was repeated for different physical layer (PHY) data rates. For each setup, we collected traces for two distinct packet transmission rates. The transmission rate is controlled by adjusting the time interval t between packets. In setup A we collected traces at 500 Kbps and 1024 Kbps, while in Setup B we collected

traces at 750 Kbps and 900 Kbps. For ease of notation, we prefer to label the traces by their PHY data rate and a single number.

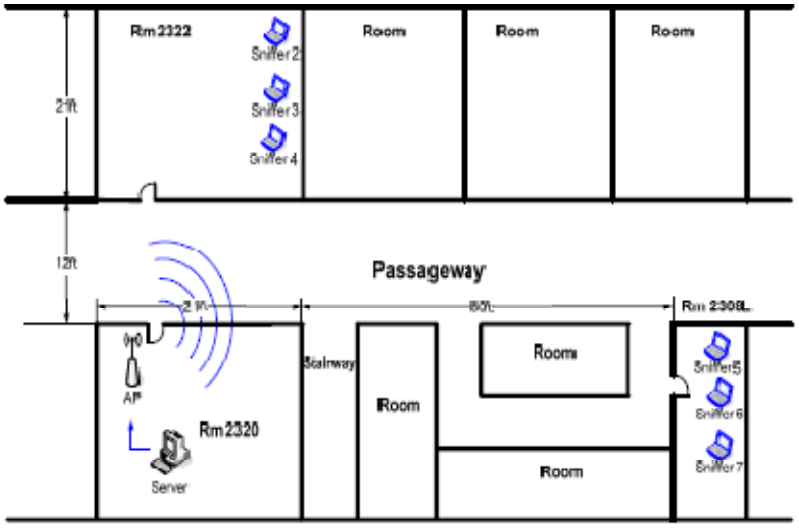
For each packet, in addition to its header and payload information, the following three additional parameters were also logged at the receivers:

- Background Traffic (BT): A four byte number representing the total number of background packets observed between two trace packets;
- Signal Strength (S) for the received packet: A one byte number representing the signal strength in dBm;

Silence Value (N) for the received packet: A one byte number which can be said to be representing the noise + interference strength in dBm.



(a) Setup A



(b) Setup B

Figure 2: 802.11 Trace Collection Setup

Collection methodology

We generated bit error traces at all bitrates supported by the standard under various settings of an 802.11b network. Network traffic at many constant bitrates was transmitted over the wireless medium. All the bit error traces were collected at the clients by modifying the wireless device drivers. More specifically, the clients were Linux boxes using DLink DWL-650 wireless PC-Cards with prism2 device drivers. The modified client device drivers passed all the packets to a (link layer) raw socket. Thus the traces collected at the clients included successful (i.e., packets with no errors) and unsuccessful (i.e., packets failing the 802.11 MAC layer checksum) transmissions. These link layer traces were copied in the kernel buffer space from where an application thread periodically concatenated them in the user buffer space. To capture packets at high transmission rates, packet dissectors were implemented inside the device drivers. These packet dissectors ensured that only packets pertinent to our wireless experiment were processed, while all other packets were dropped.

Hole/error/limitations

The data set does not have any information about the packets which have been totally lost. Only those packets are captured which are received completely (although may be in error).

Anonymization technique (if applicable)

N/A

Data format

The dataset consists of three different setups. For each setup, several traces are collected. Name of the files are meaningful. For example, trace_2mbps_500kbps_accros_sniffer1 refers to the trace collected at 2Mbps physical data rate 500Kbps MAC layer data rate and the packet is logged at receiver named sniffer1.

Each file represents one complete trace. These traces are in pcap format. Complete frames are present in the trace files. Each frame contains 1000 bytes of data. The data bytes are hexadecimal “AA” (binary 10101010). Actual bit in error can be computed by XORing received byte to “AA”.